

## Prognostic significance of abnormal hematological parameters in severe traumatic brain injury requiring decompressive craniectomy

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**OBJECTIVE** Hematological abnormalities after severe traumatic brain injury (TBI) are common, and are associated with a poor outcome. Whether these abnormalities offer additional prognostic significance over and beyond validated TBI prognostic models is uncertain.

**METHODS** This retrospective cohort study compared the ability of admission hematological abnormalities to that of the IMPACT (International Mission for Prognosis and Analysis of Clinical Trials) prognostic model to predict 18-month neurological outcome of 388 patients who required a decompressive craniectomy after severe TBI, between 2004 and 2016, in Western Australia. Area under the receiver operating characteristic (AUROC) curve was used to assess predictors' ability to discriminate between patients with and without an unfavorable outcome of death, vegetative state, or severe disability.

**RESULTS** Of the 388 patients included in the study, 151 (38.9%) had an unfavorable outcome at 18 months after decompressive craniectomy for severe TBI. Abnormalities in admission hemoglobin (AUROC 0.594,  $p = 0.002$ ), plasma glucose (AUROC 0.592,  $p = 0.002$ ), fibrinogen (AUROC 0.563,  $p = 0.036$ ), international normalized ratio (INR; AUROC 0.645,  $p = 0.001$ ), activated partial thromboplastin time (AUROC 0.564,  $p = 0.033$ ), and disseminated intravascular coagulation score (AUROC 0.623,  $p = 0.001$ ) were all associated with a higher risk of unfavorable outcome at 18 months after severe TBI. As a marker of inflammation, neutrophil to lymphocyte ratio was not significantly associated with the risk of unfavorable outcome (AUROC 0.500,  $p = 0.998$ ). However, none of these parameters, in addition to the platelet count, were significantly associated with an unfavorable outcome after adjusting for the IMPACT predicted risk (odds ratio [OR] per 10% increment in risk 2.473, 95% confidence interval [CI] 2.061–2.967;  $p = 0.001$ ). After excluding 8 patients (2.1%) who were treated with warfarin prior to the injury, there was a suggestion that INR was associated with some additional prognostic significance (OR 3.183, 95% CI 0.856–11.833;  $p = 0.084$ ) after adjusting for the IMPACT predicted risk.

**CONCLUSIONS** In isolation, INR was the best hematological prognostic parameter in severe TBI requiring decompressive craniectomy, especially when patients treated with warfarin were excluded. However, the prognostic significance of admission hematological abnormalities was mostly captured by the IMPACT prognostic model, such that they did not offer any additional prognostic information beyond the IMPACT predicted risk. These results suggest that new prognostic factors for TBI should be evaluated in conjunction with predicted risks of a comprehensive prognostic model that has been validated, such as the IMPACT prognostic model.

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**KEYWORDS** head injury; hemorrhage; prognosis; surgery; trauma; traumatic brain injury

**A**BNORMAL hematological parameters are common after severe traumatic brain injury (TBI). A systematic review of 34 studies over a 40-year period reported an overall incidence of coagulopathy in as many as 60% of patients with severe TBI.<sup>10</sup> More importantly,

the presence of coagulopathy after TBI was reported to be associated with a dramatic increase in mortality compared to those without coagulopathy (46% vs 7%).<sup>6,7</sup> In addition to coagulopathy, a reduced hemoglobin concentration and hyperglycemia are also common after TBI, and associated

**ABBREVIATIONS** aPTT = activated partial thromboplastin time; AUROC = area under the receiver operating characteristic; CI = confidence interval; CRASH = Corticosteroid Randomization After Significant Head Injury; DIC = disseminated intravascular coagulation; GCS = Glasgow Coma Scale; GOS = Glasgow Outcome Scale; IMPACT = International Mission for Prognosis and Analysis of Clinical Trials; INR = international normalized ratio; NLR = neutrophil to lymphocyte ratio; OR = odds ratio; SAH = subarachnoid hemorrhage; TBI = traumatic brain injury.

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with an increased mortality rate, possibly through inducing secondary brain injury.<sup>16</sup> Recent studies have suggested that, as a marker of inflammation, an increase in neutrophil to lymphocyte ratio (NLR) is common after a variety of acute neurosurgical emergencies, and is associated with a worse mortality outcome after TBI, intracerebral hemorrhage, and subarachnoid hemorrhage (SAH).<sup>5,17,19</sup>

Recent improvements in statistical techniques, combined with access to large clinical data sets, have enabled investigators to develop sophisticated web-based outcome prediction models. The CRASH (Corticosteroid Randomization After Significant Head injury) and IMPACT (International Mission for Prognosis and Analysis of Clinical Trials) models incorporate a number of prognostic factors in order to provide a prediction of unfavorable neurological outcome at 6 months.<sup>14,16</sup> A number of studies have subsequently externally validated these models.<sup>8,11,12</sup> Although these two validated prognostic models have excellent predictive ability to discriminate between patients with and without long-term unfavorable outcome (with area under the receiver operating characteristic [AUROC] curve > 0.80), it is still possible to further improve their predictive accuracy by incorporating important prognostic factors, such as abnormal hematological parameters, that are not considered by these two models.

We hypothesized that in patients with severe TBI, abnormal admission hematological parameters, including the coagulation profile, platelet count, and NLR, can be useful to discriminate between patients with and without 18-month unfavorable neurological outcome after severe TBI requiring decompressive craniectomy. In this retrospective cohort study, we compared the prognostic significance of abnormal hematological parameters to the IMPACT prognostic model. And specifically, we also assessed whether the ability of the IMPACT prognostic model to predict long-term neurological outcome could be further improved by combining it with the hematological parameters. We used the IMPACT model instead of the CRASH model for comparison, because we have previously demonstrated that the IMPACT model was better calibrated than the CRASH model in this group of patients.<sup>11</sup>

## Methods

### Database Review

After obtaining approval for the study from the Royal Perth Hospital Clinical Quality and Safety Unit, the admission laboratory data of patients who underwent a decompressive craniectomy after severe TBI requiring decompressive craniectomy, between 2004 and 2016 at the two tertiary neurosurgical referral centers in Western Australia, were retrieved and merged with the existing statewide neurosurgical administrative database. The two neurosurgical study centers are the only adult neurotrauma centers in the state of Western Australia and serve a population of approximately 2.1 million.

The database contains all the clinical (age, pupil reactivity, motor score, hypoxia, and hypotension), laboratory (plasma glucose, hemoglobin concentrations), and radiological information (presence of epidural hematoma and

SAH, Marshall CT grading) on admission to the study centers that was needed to calculate the predicted risks of unfavorable outcome at 6 months using the IMPACT model (including full laboratory and CT brain data). The database also recorded the 18-month neurological outcomes of the patients according to the Glasgow Outcome Scale (GOS). Similar to all TBI outcome studies, GOS scores between 1 and 3—corresponding to death, vegetative state, or severe disability, respectively—were considered as an unfavorable outcome in this study. We had previously validated the accuracy of the IMPACT model in predicting the neurological outcome at 18 months after severe TBI requiring decompressive craniectomy.<sup>12</sup> The IMPACT prognostic model calculator is freely available on the Internet ([www.tbi-impact.org](http://www.tbi-impact.org)).

### Prognostic Variables

The hematological parameters assessed in this study included international normalized ratio (INR), activated partial thromboplastin time (aPTT), fibrinogen, platelet count, total white cell count, and NLR, all obtained on hospital admission after TBI. Because a combination of abnormal coagulation parameters, such as the disseminated intravascular coagulation (DIC) score,<sup>18</sup> may have a stronger prognostic significance than one abnormality alone, we also assessed the prognostic value of the DIC score with and without adjusting for the IMPACT predicted risk. In this study, only platelet count, prothrombin time, and fibrinogen concentration were used to calculate the DIC score because D-dimers or fibrin degradation products were not routinely measured in patients with severe TBI in the study centers.

### Statistical Analyses

Categorical and skewed continuous data were analyzed by chi-square and Mann-Whitney tests, respectively. AUROC curve was used to define the ability of different hematological parameters and the predicted risks of the IMPACT model to discriminate between patients with and without unfavorable outcome at 18 months after severe TBI. The difference in AUROC derived from the same cases was estimated according to the method suggested by Hanley and McNeil.<sup>9</sup> To assess whether the hematological parameters would add prognostic significance to the IMPACT model, we analyzed these hematological parameters simultaneously with the IMPACT predicted risk of unfavorable outcome in a multivariate logistic regression model.

As a sensitivity analysis, we repeated the multivariate analysis after excluding patients who were treated with warfarin prior to severe TBI, because therapeutic anticoagulant may confound the effect of brain injury on the coagulation parameters. We also assessed whether INR had a “dose-related” or proportional relationship with the severity of TBI—measured by the IMPACT predicted risks of unfavorable outcome—by Pearson’s correlation coefficient and linear regression in a scatterplot. All statistical analyses were conducted by IBM SPSS for Windows (version 24.0, IBM Corp.) and MedCalc for Windows (version 12.5). A 2-tailed  $\alpha$ -error < 5% was considered significant.

**TABLE 1. Difference in characteristics between those with favorable and unfavorable outcome at 18 months after decompressive craniectomy for severe TBI**

| Variables                                                       | All Patients (n = 388) | Outcome               |                     | p Value* |
|-----------------------------------------------------------------|------------------------|-----------------------|---------------------|----------|
|                                                                 |                        | Unfavorable (n = 151) | Favorable (n = 237) |          |
| Med age (IQR), yrs                                              | 33 (22–49)             | 37 (23–54)            | 30 (21–45)          | 0.005    |
| Males, n (%)                                                    | 310 (80)               | 118 (78)              | 192 (81)            | 0.492    |
| Med GCS score (IQR)                                             | 8 (5–11)               | 5 (4–8)               | 9 (7–12)            | 0.001    |
| Pupil reactivity, n (%)                                         |                        |                       |                     | 0.001    |
| None                                                            | 29 (8)                 | 26 (17)               | 3 (1)               |          |
| 1                                                               | 63 (16)                | 37 (25)               | 26 (11)             |          |
| 2                                                               | 296 (76)               | 88 (58)               | 208 (88)            |          |
| CT brain findings, n (%)                                        |                        |                       |                     |          |
| tSAH                                                            | 361 (93)               | 146 (97)              | 217 (92)            | 0.045    |
| Midline shift                                                   | 330 (85)               | 134 (89)              | 196 (83)            | 0.104    |
| Epidural hematoma                                               | 24 (6)                 | 5 (3)                 | 19 (8)              | 0.061    |
| Unevacuated hematoma                                            | 159 (41)               | 64 (42)               | 95 (40)             | 0.653    |
| Effaced basal cistern                                           | 96 (25)                | 56 (37)               | 40 (17)             | 0.001    |
| Petechial hemorrhage                                            | 368 (95)               | 143 (95)              | 225 (95)            | 0.919    |
| Major extracranial injury, n (%)                                | 140 (36)               | 68 (45)               | 72 (30)             | 0.003    |
| Bilateral craniectomy, n (%)                                    | 152 (39)               | 57 (38)               | 95 (40)             | 0.646    |
| Med length of ICU stay (IQR), days                              | 10 (5–14)              | 10 (4–14)             | 10 (6–13)           | 0.824    |
| Med length of ward stay (IQR), days                             | 23 (13–45)             | 34 (0–103)            | 20 (15–31)          | 0.042    |
| Med IMPACT predicted unfavorable outcome risk at 6 mos (IQR), % | 44 (31–64)             | 68 (53–81)            | 34 (26–47)          | 0.001    |
| Prior warfarin treatment, n (%)                                 | 8 (2)                  | 3 (2)                 | 5 (2)               | 0.934    |
| Med admission biochem & hematological parameters (IQR)          |                        |                       |                     |          |
| Glucose, mmol/L                                                 | 8 (6–10)               | 9 (7–10)              | 8 (6–9)             | 0.002    |
| Hemoglobin, g/dl                                                | 14 (12–15)             | 13 (12–15)            | 14 (12–15)          | 0.001    |
| Total white blood cells, $\times 10^9/L$                        | 14 (11–19)             | 14 (10–19)            | 14 (11–19)          | 0.711    |
| NLR                                                             | 6 (2–12)               | 6 (3–11)              | 6 (2–12)            | 0.996    |
| Platelets, $\times 10^9/L$                                      | 243 (197–302)          | 240 (181–290)         | 243 (206–305)       | 0.091    |
| Fibrinogen, g/L                                                 | 3 (2–3)                | 2 (2–3)               | 3 (2–3)             | 0.035    |
| INR                                                             | 1.1 (1–1.3)            | 1.2 (1.1–1.4)         | 1.1 (1.0–1.2)       | 0.001    |
| aPTT, sec                                                       | 30 (27–33)             | 31 (27–35)            | 30 (27–32)          | 0.031    |
| DIC score†                                                      | 0 (0–1)                | 0 (0–1)               | 0 (0–0)             | 0.001    |

Biochem = biochemical; ICU = intensive care unit; IQR = interquartile range; med = median; tSAH = traumatic SAH.

\* p values generated by chi-square or Mann-Whitney tests.

† Only patients without previous warfarin treatment were included and the score associated with D-dimers or fibrin degradation products was not included.

## Results

Of the 388 patients included in the study, 151 (38.9%, 95% confidence interval [CI] 34.2%–43.9%) had an unfavorable outcome at 18 months after decompressive craniectomy for severe TBI. The patients were predominantly young (with a median age of 33 years) and male (80%), and approximately one-third of them required bilateral craniectomy (39%) and had extracranial injuries (36%). The patient characteristics and hematological parameters for the whole cohort, and those with and without unfavorable outcomes, are described in Table 1.

Abnormalities in admission hemoglobin (AUROC 0.594, 95% CI 0.535–0.653;  $p = 0.002$ ), plasma glucose (AUROC 0.592, 95% CI 0.532–0.651;  $p = 0.002$ ), fibrinogen (0.563, 95% CI 0.505–0.622;  $p = 0.036$ ), INR (AUROC

0.645, 95% CI 0.588–0.702;  $p = 0.001$ ), aPTT (AUROC 0.564, 95% CI 0.504–0.625;  $p = 0.033$ ), and the DIC score (AUROC 0.623, 95% CI 0.564–0.682;  $p = 0.001$ ) were all associated with a higher risk of unfavorable outcome at 18 months after severe TBI than those with a favorable outcome (Table 2). Platelet count (AUROC 0.447, 95% CI 0.387–0.507;  $p = 0.078$ ) and markers of inflammation, including the NLR (AUROC 0.500, 95% CI 0.442–0.559;  $p = 0.998$ ) and total white blood cell count (AUROC 0.507, 95% CI 0.447–0.567;  $p = 0.816$ ), were not significantly associated with the risk of unfavorable outcome at 18 months after decompressive craniectomy. In the multivariate analysis, none of the hematological parameters were significantly associated with an unfavorable outcome after adjusting for the IMPACT predicted risk (odds ratio [OR] per 10% increment in risk 2.473, 95% CI 2.061–2.967;  $p = 0.001$ ; Table 3).

**TABLE 2. AUROC curve for different admission hematological parameters, as compared to IMPACT predicted risks, to discriminate between those with favorable and unfavorable outcome at 18 months after decompressive craniectomy for severe TBI**

| Variables                                | AUROC (95% CI)      | p Value      |
|------------------------------------------|---------------------|--------------|
| Hemoglobin, g/dl                         | 0.594 (0.535–0.653) | <b>0.002</b> |
| Total white blood cells, $\times 10^9/L$ | 0.507 (0.447–0.567) | 0.816        |
| NLR                                      | 0.500 (0.442–0.559) | 0.998        |
| Platelets, $\times 10^9/L$               | 0.447 (0.387–0.507) | 0.078        |
| Fibrinogen, g/L                          | 0.563 (0.505–0.622) | <b>0.036</b> |
| INR*                                     | 0.645 (0.588–0.702) | <b>0.001</b> |
| aPTT, sec                                | 0.564 (0.504–0.625) | <b>0.033</b> |
| DIC score†                               | 0.623 (0.564–0.682) | <b>0.001</b> |
| Glucose, mmol/L                          | 0.592 (0.532–0.651) | <b>0.002</b> |

Boldface values indicate statistical significance.

\* AUROC = 0.652, 95% CI 0.595–0.709,  $p = 0.001$ , after excluding 8 patients who were treated with warfarin.

† Only patients without warfarin treatment were included and the score associated with D-dimers or fibrin degradation products was not included.

In the sensitivity analysis, after excluding 8 patients (2.1%) who were treated with warfarin prior to severe TBI, there was a small increase in the ability of INR in isolation to predict an unfavorable outcome (AUROC 0.651), but its accuracy was still far inferior to the IMPACT predicted risks (AUROC 0.870,  $p < 0.001$ , when comparing the two AUROC curves; Fig. 1). The DIC score was not statistically significantly (OR 1.402 per point increment, 95% CI 0.991–1.983;  $p = 0.056$ ) associated with the risk of unfavorable outcome at 18 months when simultaneously adjusted for the IMPACT predicted risk (OR 2.381 per 10% predicted risk increment, 95% CI 2.005–2.827;  $p = 0.001$ ).

The scatterplot between INR and IMPACT predicted risks showed that INR was weakly related to the severity of TBI (Pearson's correlation coefficient  $r = 0.24$ ,  $p < 0.001$ ; Fig. 2). In the multivariate analysis, after excluding the patients treated with warfarin prior to admission, there was a suggestion that INR (OR 3.183, 95% CI 0.856–11.833;  $p = 0.084$ ) may add some degree of prognostic significance to the IMPACT predicted risks, albeit this result did not reach the predefined statistical significance.

## Discussion

This study showed that coagulation parameters, including the INR, aPTT, fibrinogen, and DIC score, were more likely to be abnormal among those who had an unfavorable long-term neurological outcome after decompressive craniectomy. Among all the hematological parameters considered, INR had a weak proportional association with the severity of TBI, and was the best hematological predictor in isolation, in predicting unfavorable outcome (AUROC 0.651). However, none of the hematological parameters, including all coagulation parameters and NLR, were as good as the IMPACT prediction model, nor could they supplement the IMPACT model in predicting long-term neurological outcome. These results have some significance and require further discussion.

**TABLE 3. Ability of different hematological parameters to predict unfavorable outcome at 18 months after decompressive craniectomy while adjusting for the IMPACT predicted risks of unfavorable outcome**

| Variables                                            | OR (95% CI)         | p Value |
|------------------------------------------------------|---------------------|---------|
| IMPACT predicted risk (OR per 10% increment in risk) | 2.473 (2.061–2.967) | 0.001   |
| Hemoglobin, g/dl                                     | 1.005 (0.986–1.024) | 0.619   |
| Total white blood cells, $\times 10^9/L$             | 1.003 (0.955–1.053) | 0.906   |
| NLR                                                  | 1.003 (0.972–1.035) | 0.870   |
| Platelets, $\times 10^9/L$                           | 0.999 (0.995–1.003) | 0.537   |
| Fibrinogen, g/L                                      | 0.938 (0.714–1.232) | 0.644   |
| INR*                                                 | 1.276 (0.649–2.510) | 0.479   |
| aPTT, sec                                            | 0.986 (0.943–1.031) | 0.535   |
| Glucose, mmol/L                                      | 0.962 (0.873–1.060) | 0.435   |

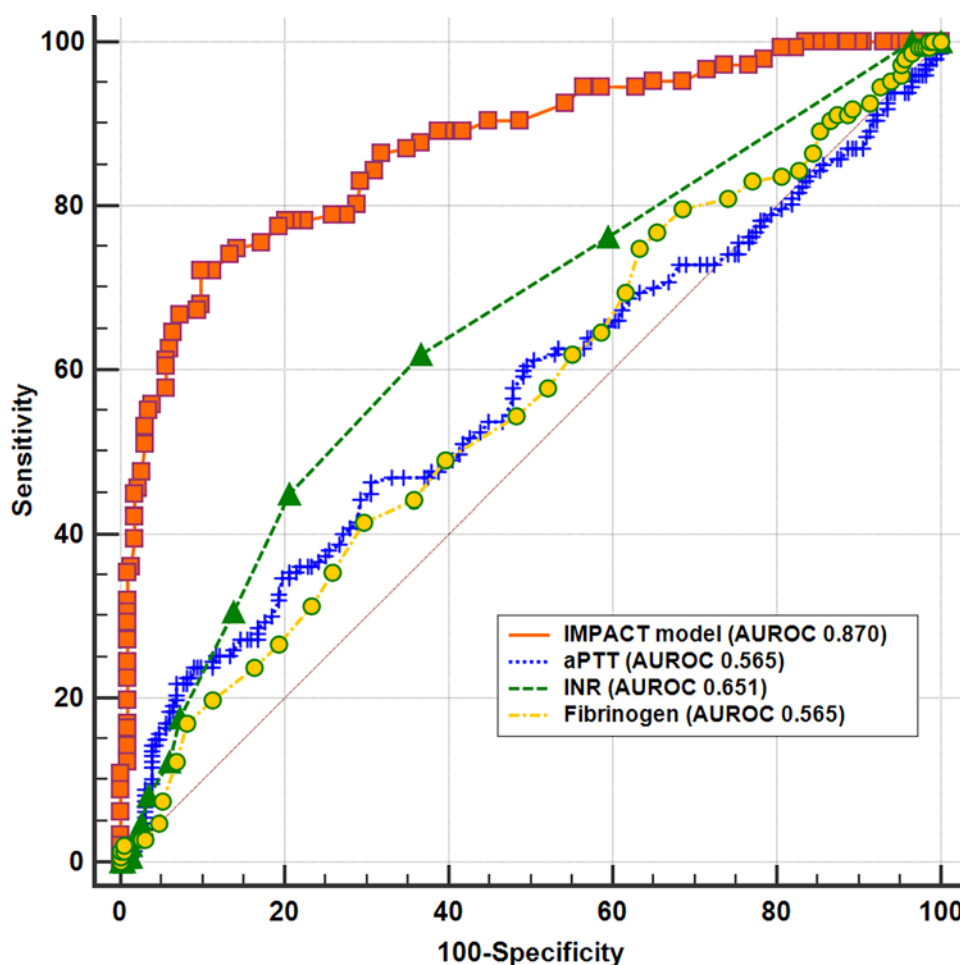
Nagelkerke  $R^2$  and Hosmer-Lemeshow chi-square of the final model were 0.530 and 11.4 ( $p = 0.180$ ), respectively.

\* Excluding 8 patients treated with warfarin showed that there was a suggestion that INR might add prognostic significance to the IMPACT predicted risk (OR 3.183 per 10% risk increment, 95% CI 0.856–11.833;  $p = 0.084$ ). The DIC score (without including the score associated with D-dimers or fibrin degradation products) was not statistically significantly (OR 1.402 per score increment, 95% CI 0.991–1.983;  $p = 0.056$ ) associated with the risk of unfavorable outcome at 18 months after surgery when simultaneously adjusted for the IMPACT predicted risk (OR 2.381 per 10% risk increment, 95% CI 2.005–2.827;  $p = 0.001$ ).

Although previous studies have showed that many hematological parameters, including the coagulation profiles and NLR, were important prognostic factors in TBI and other neurosurgical emergencies, most, if not all, of these studies only assessed these hematological parameters after adjusting for a few clinical parameters such as age, Glasgow Coma Scale (GCS) score, and pupils' reactivity to light,<sup>5–7</sup> without directly comparing with a validated TBI prognostic model derived from a multicenter study. We found that at least three coagulation parameters had some prognostic significance on their own in the univariate analyses (Table 2). Nevertheless, none of these coagulation parameters offered additional prognostic significance over and above the IMPACT prognostic model. It is possible that abnormal coagulation parameters are just surrogate markers of the severity of the underlying TBI due to cerebral contusions and edema (with midline shift, basal cistern effacement, SAH, and nonreactive pupils), all part of the IMPACT prognostic model. If abnormal coagulation parameters are mainly "bystanders" (or confounders through their association with other prognostic factors) of severe TBI, it is not surprising that they do not add prognostic significance to the IMPACT prognostic model in the multivariate analysis.

Our results showed that INR was the best hematological prognostic marker in isolation, and it had a direct positive relationship with the IMPACT predicted risks of unfavorable outcome. This finding is largely consistent with a variety of mechanisms proposed to explain why coagulopathy is common after severe TBI, including hemodilution, reduced tissue perfusion, and disruption of the blood-brain barrier leading to tissue factor release and consumption





**FIG. 1.** AUROC curves of the three coagulation parameters in discriminating between those with and without unfavorable outcome at 18 months after decompressive craniectomy for severe TBI compared to the IMPACT predicted risk ( $n = 380$ , after excluding 8 patients treated with warfarin prior to injury). Figure is available in color online only.

of coagulation factors.<sup>6,13</sup> The relatively weak association between INR and the IMPACT predicted risk ( $r = 0.24$ ) is likely related to the fact that the IMPACT risk contains a vast amount of prognostic information over and beyond what the INR can provide, including the location and operability of the intracranial hematoma and age of the patients. It is, however, important to keep the prognosis and treatment role of the INR or coagulopathy in severe TBI distinct. Although the INR may not add prognostic value beyond the IMPACT model, this does not mean that the INR should not be considered in the clinical management of patients with severe TBI. Recent studies showed that mild coagulopathy and a slightly raised INR (e.g.,  $< 1.6$ ) do not correlate with hypocoagulability on a viscoelastic test and risk of cerebral hemorrhage following the placement of ventriculostomy, respectively.<sup>3,15</sup> Nonetheless, a recent systematic review of 19 studies suggested that a raised INR was associated with an increased risk of progressive hemorrhagic brain injury after TBI.<sup>20</sup> Whether correction of an abnormal INR and other coagulation parameters with blood products such as fresh frozen plasma would improve outcome is highly contentious, with conflicting results from different retrospective studies.<sup>1,4</sup> Similar to

the situation in patients who develop hemorrhagic stroke while receiving aspirin therapy,<sup>2</sup> a randomized controlled trial is needed to assess whether aggressive correction of coagulopathy in patients with severe TBI can improve patient-centered outcomes.

Finally, we would like to acknowledge the limitations of this study. First, this study only included patients with severe TBI requiring decompressive craniectomy. It is possible that coagulopathy may have a stronger prognostic significance in mild or moderate TBI, in which abnormal CT brain findings are less common and hence less relevant.<sup>10</sup> In addition, we did not have the precise data on the duration between injury and hospital admission, and also prior treatment, including possible blood transfusion, before the hematological blood tests in our patients. These factors may have confounded our results. Second, we only assessed long-term neurological functional outcomes, without considering morbidity events such as venous thromboembolism and bleeding complications that are also important from healthcare utilization and patients' perspectives. Third, while this is a reasonably large series of patients with severe TBI and long-term outcome data, it remains possible that our study is underpowered to detect a small,

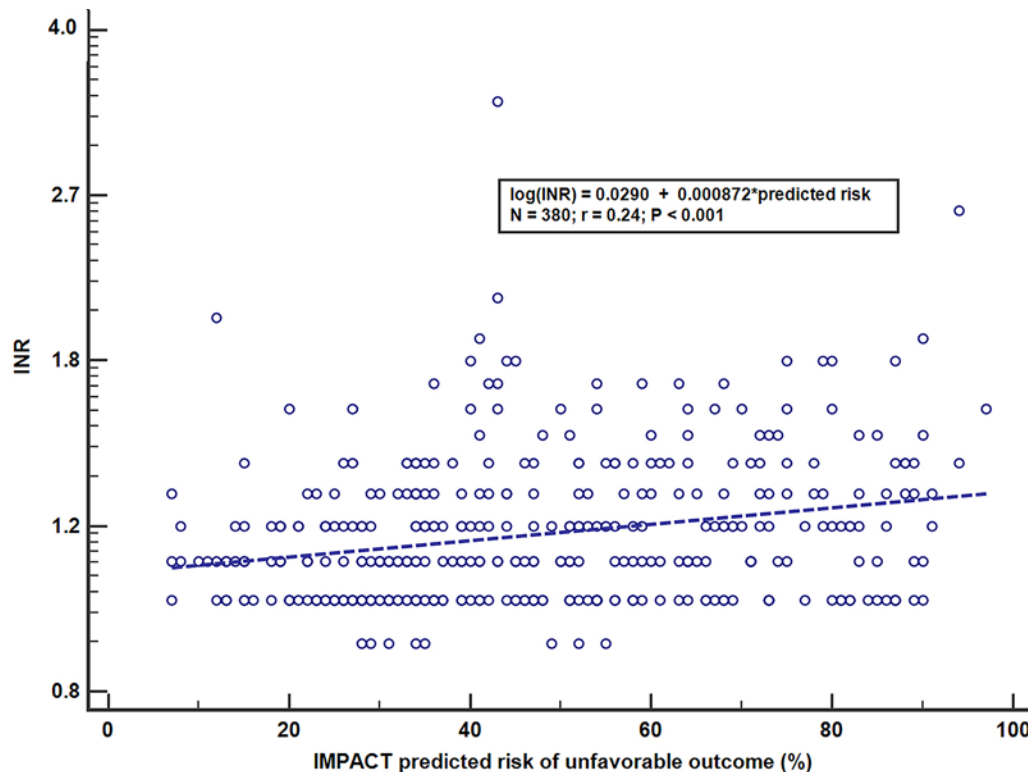


FIG. 2. Scatterplot showing the relationship between admission INR and the IMPACT predicted risks of unfavorable outcome at 6 months. Figure is available in color online only.

but clinically relevant, prognostic significance of the INR that is independent of the IMPACT predicted risk. As such, large prospective studies are needed to confirm the utility of incorporating the INR into any new TBI prognostic models.<sup>21</sup>

## Conclusions

In isolation, the INR was the best hematological prognostic parameter in severe TBI requiring decompressive craniectomy, especially when patients treated with warfarin were excluded. However, its prognostic significance was still inferior to the IMPACT prognostic model, and none of the admission hematological abnormalities appeared to add prognostic information beyond what was already captured by the IMPACT prognostic model. Our results suggest that new prognostic factors for TBI should be evaluated in conjunction with predicted risks of a comprehensive prognostic model that has been validated, such as the IMPACT prognostic model. Clinicians should also keep the therapeutic and prognostic roles of hematological abnormalities distinct. Whether aggressive correction of hematological abnormalities by transfusion will improve long-term neurological outcome after severe TBI requires further assessment by an adequately powered randomized controlled trial.

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## Disclosures

The authors report no conflict of interest concerning the materials or methods used in this study or the findings specified in this paper.

## Author Contributions

Conception and design: all authors. Acquisition of data: Corbett, Honeybul. Analysis and interpretation of data: all authors. Drafting the article: all authors. Critically revising the article: all authors. Reviewed submitted version of manuscript: all authors. Approved the final version of the manuscript on behalf of all authors: Ho. Statistical analysis: Ho. Administrative/technical/material support: Honeybul. Study supervision: Ho, Honeybul.

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